

V4 model diagnostic verification

Six models · SPY, AAPL, MSFT · four windows · 1.5-year monthly study

What this report is

- Return-based specification tests for each model × company × regime cell. It does not price options and does not use LSM RMSE or MAE.
- One table per regime. Rows are GBM, Modified GBM, GARCH, Heston, Merton, GARCH-Merton, Heston-Merton. Columns are SPY, AAPL, MSFT, AMZN, each with JB, Q(20), Q²(20), and ARCH-LM.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

Sample and residuals

- Daily $R_t = \ln(S_t / S_{t-1})$ inside that regime window only. No lookback from outside the window.
- GBM: $(R_t - \hat{\mu}) / \hat{\sigma}$. Modified GBM: split-normal magnitude residual, up vs down. GARCH: GARCH(1,1) z_t . Merton: jump-adjusted diffusion σ . GARCH-Merton / Heston-Merton: residuals after the 3σ jump filter. Heston: filtered variance path from returns.
- A diagnostic is N/A if the residual series is not well-defined or the sample is too short for that test.

Tests

- JB: usual Jarque-Bera test of normality of the standardized residuals.
- Q(20): Ljung-Box test of residual autocorrelation through lag 20.
- Q²(20): Ljung-Box test of autocorrelation in squared standardized residuals through lag 20.
- ARCH-LM: Engle (1982) test for remaining ARCH effects, 5 lags. Statistic with p-value in brackets.

Table D1 · 2008-2009 Crisis · model diagnostics

2008-08-01 → 2009-07-31 · n = 252 daily observations · statistic [p-value]

Model	SPY				AAPL				MSFT					AMZN			
Model	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	
GBM	41.86 [<0.003]	38.16 [0.008]	74.38 [<0.001]	0.255 [<0.001]	11.07 [<0.003]	38.31 [0.008]	83.66 [<0.001]	12.08 [0.031]	16.95 [<0.003]	37.03 [0.012]	36.98 [0.012]	8.08 [0.152]	29.16 [<0.001]	28.49 [0.098]	5.77 [<0.001]	19.27 [0.002]	
Modified GBM	656.44 [<0.002]	5.58 [0.180]	1.77 [<0.001]	4.20 [<0.001]	77.28 [<0.001]	16.50 [0.685]	5.64 [<0.001]	3.76 [0.585]	92.82 [<0.003]	33.24 [0.032]	14.28 [0.816]	1.34 [0.931]	106.27 [<0.002]	17.18 [0.130]	67.11 [<0.001]	12.20 [0.032]	
GARCH	8.05 [0.018]	18.62 [0.547]	19.71 [0.476]	4.44 [0.488]	30.73 [<0.001]	18.03 [0.585]	17.81 [0.600]	1.23 [0.942]	59.22 [<0.001]	22.14 [0.333]	11.25 [0.939]	1.41 [0.923]	23.76 [<0.001]	18.60 [0.548]	10.15 [0.965]	2.42 [0.789]	
Heston	11.23 [0.004]	19.51 [0.489]	29.28 [0.082]	20.81 [<0.001]	13.70 [0.001]	20.43 [0.432]	26.59 [0.147]	7.12 [0.212]	10.12 [0.006]	16.34 [0.696]	22.67 [0.305]	6.87 [0.231]	13.17 [0.001]	17.61 [0.613]	30.55 [0.061]	16.11 [0.007]	
Merton	41.86 [<0.003]	38.16 [0.008]	75.30 [<0.001]	0.280 [<0.001]	11.07 [<0.003]	38.31 [0.008]	83.35 [<0.001]	12.03 [0.031]	16.95 [<0.003]	37.03 [0.012]	35.75 [0.016]	7.64 [0.177]	29.16 [<0.001]	28.49 [0.098]	3.57 [<0.001]	18.79 [0.002]	
GARCH-Me	2.47 [0.291]	16.45 [0.688]	19.63 [0.481]	6.28 [0.280]	0.39 [0.821]	19.99 [0.459]	18.65 [0.545]	4.26 [0.512]	3.19 [0.203]	10.80 [0.951]	18.67 [0.543]	4.40 [0.493]	1.19 [0.551]	14.05 [0.828]	28.08 [0.107]	10.27 [0.068]	
Heston-Me	11.35 [0.003]	18.15 [0.577]	28.45 [0.099]	17.57 [0.004]	4.67 [<0.002]	23.04 [0.287]	29.15 [0.085]	8.75 [0.119]	11.32 [0.003]	13.29 [0.865]	23.77 [0.252]	9.29 [0.098]	14.03 [<0.001]	13.50 [0.855]	30.42 [0.063]	19.04 [0.002]	

Notes: This table reports residual diagnostics for GBM, Modified GBM, GARCH, Heston, Merton, GARCH-Merton, Heston-Merton fitted to daily continuously compounded returns $R_t = \ln(S_t / S_{t-1})$ of SPY, AAPL, MSFT, AMZN. The sample is August 1, 2008 to July 31, 2009 (252 trading days). Each model is estimated from that regime and company only. Jump models use standardized residuals after the 3σ jump filter. JB is the usual Jarque-Bera normality test for the standardized residuals. Q(20) is the Ljung-Box portmanteau test for serial correlation in the standardized residuals up to 20 lags; Q²(20) is the same test on the squared standardized residuals. ARCH-LM is Engle's (1982) test for remaining ARCH effects with 5 lags. The test statistic is reported with the p-value in brackets. N/A means the diagnostic is not applicable or the residual series is not well-defined. These tests check model adequacy on returns; they are not option pricing RMSE or MAE.

Table D2 · 2013-2014 Normal · model diagnostics

2014-01-01 → 2014-12-31 · n = 253 daily observations · statistic [p-value]

Model	SPY				AAPL				MSFT					AMZN			
Model	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	
GBM	24.08 [<0.001]	14.15 [0.823]	55.08 [<0.001]	22.71 [<0.001]	107.48 [<0.001]	24.03 [0.241]	3.79 [1.000]	1.79 [0.878]	14.75 [<0.001]	25.67 [0.177]	15.11 [0.770]	8.81 [0.117]	354.16 [<0.001]	19.51 [0.489]	7.29 [0.996]	2.39 [0.793]	
Modified GBM	56.74 [<0.001]	22.26 [0.327]	32.61 [0.037]	16.88 [0.006]	15.19 [<0.001]	15.54 [0.745]	2.55 [1.000]	1.47 [0.916]	25.43 [<0.001]	17.47 [0.622]	16.10 [0.711]	11.35 [0.046]	607.03 [<0.001]	19.55 [0.486]	13.13 [0.872]	7.59 [0.180]	
GARCH	33.44 [<0.001]	11.37 [0.936]	9.39 [0.978]	2.45 [0.784]	147.09 [<0.001]	23.72 [0.255]	3.15 [1.000]	1.14 [0.950]	10.41 [0.005]	25.23 [0.193]	12.30 [0.905]	2.25 [0.815]	307.62 [<0.001]	22.75 [0.301]	8.56 [0.988]	0.76 [0.979]	
Heston	14.16 [<0.001]	19.57 [0.975]	11.45 [0.934]	5.00 [0.416]	12.35 [0.002]	22.79 [0.299]	20.25 [0.442]	3.71 [0.592]	15.85 [<0.001]	26.16 [0.161]	18.21 [0.574]	5.74 [0.332]	8.51 [0.014]	18.29 [0.568]	42.26 [0.003]	12.19 [0.032]	
Merton	24.08 [<0.001]	14.15 [0.823]	55.08 [<0.001]	22.71 [<0.001]	107.48 [<0.001]	24.03 [0.241]	3.78 [1.000]	1.76 [0.882]	14.75 [<0.001]	25.67 [0.177]	15.08 [0.772]	8.81 [0.117]	354.16 [<0.001]	19.51 [0.489]	7.04 [0.997]	2.17 [0.825]	
GARCH-Merton	10.68 [0.005]	11.91 [0.919]	26.36 [0.154]	11.13 [0.049]	4.39 [0.111]	16.73 [0.671]	9.20 [0.980]	1.49 [0.915]	0.51 [0.777]	29.74 [0.074]	19.64 [0.481]	9.10 [0.105]	7.66 [0.022]	18.10 [0.581]	28.59 [0.096]	6.46 [0.264]	
Heston-Merton	44.05 [<0.001]	10.38 [0.961]	12.37 [0.903]	4.06 [0.541]	14.06 [<0.001]	22.33 [0.323]	14.08 [0.826]	6.45 [0.265]	16.85 [<0.001]	18.96 [0.524]	30.60 [0.061]	13.16 [0.022]	12.00 [0.002]	11.47 [0.933]	33.88 [0.027]	11.23 [0.047]	

Notes: This table reports residual diagnostics for GBM, Modified GBM, GARCH, Heston, Merton, GARCH-Merton, Heston-Merton fitted to daily continuously compounded returns $R_t = \ln(S_t / S_{t-1})$ of SPY, AAPL, MSFT, AMZN. The sample is January 1, 2014 to December 31, 2014 (253 trading days). Each model is estimated from that regime and company only. Jump models use standardized residuals after the 3σ jump filter. JB is the usual Jarque-Bera normality test for the standardized residuals. Q(20) is the Ljung-Box portmanteau test for serial correlation in the standardized residuals up to 20 lags; Q²(20) is the same test on the squared standardized residuals. ARCH-LM is Engle's (1982) test for remaining ARCH effects with 5 lags. The test statistic is reported with the p-value in brackets. N/A means the diagnostic is not applicable or the residual series is not well-defined. These tests check model adequacy on returns; they are not option pricing RMSE or MAE.

Table D3 · 2018-2019 Late-cycle · model diagnostics

2018-10-01 → 2019-09-30 · n = 251 daily observations · statistic [p-value]

Model	SPY				AAPL				MSFT					AMZN			
Model	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	
GBM	79.84 [<0.001]	14.66 [0.795]	1.48 [<0.001]	13.12 [0.022]	25.74 [<0.001]	15.77 [0.731]	29.59 [0.077]	9.59 [0.088]	29.09 [<0.001]	26.75 [0.142]	70.00 [<0.001]	14.68 [0.012]	24.39 [<0.001]	24.15 [0.236]	35.87 [<0.001]	32.41 [<0.001]	
Modified GBM	452.89 [<0.001]	21.21 [0.385]	44.34 [0.001]	4.42 [0.491]	36.69 [<0.001]	27.85 [0.113]	17.10 [<0.001]	20.53 [<0.001]	27.49 [<0.001]	24.66 [0.215]	39.76 [0.005]	10.28 [0.068]	27.12 [<0.001]	25.92 [0.168]	78.06 [<0.001]	21.97 [<0.001]	
GARCH	74.40 [<0.001]	19.73 [0.973]	10.17 [0.965]	2.17 [0.826]	68.09 [<0.001]	15.97 [0.718]	11.56 [0.930]	1.86 [0.868]	22.84 [<0.001]	15.66 [0.738]	15.82 [0.728]	2.02 [0.846]	8.97 [0.011]	14.56 [0.801]	19.09 [0.516]	3.62 [0.605]	
Heston	11.00 [0.004]	12.07 [0.914]	28.27 [0.103]	11.15 [0.048]	10.65 [0.005]	16.57 [0.681]	20.49 [0.427]	12.06 [0.034]	11.77 [0.003]	16.61 [0.678]	16.83 [0.664]	5.15 [0.397]	10.14 [0.006]	16.50 [0.685]	20.17 [0.447]	8.65 [0.124]	
Merton	79.84 [<0.001]	14.66 [0.795]	1.39 [<0.001]	13.03 [0.023]	25.74 [<0.001]	15.77 [0.731]	29.57 [0.077]	9.53 [0.090]	29.09 [<0.001]	26.75 [0.142]	69.88 [<0.001]	14.65 [0.012]	24.39 [<0.001]	24.15 [0.236]	34.65 [<0.001]	32.12 [<0.001]	
GARCH-Me	6.65 [0.036]	16.02 [0.715]	15.98 [0.718]	2.21 [0.820]	6.22 [0.045]	14.94 [0.780]	13.85 [0.838]	1.11 [0.953]	6.42 [0.040]	12.12 [0.912]	16.53 [0.683]	0.73 [0.981]	1.11 [0.575]	13.24 [0.867]	20.99 [0.398]	4.05 [0.543]	
Heston-Me	11.29 [0.004]	13.44 [0.858]	41.62 [0.003]	17.43 [0.004]	11.63 [0.003]	12.65 [0.892]	21.70 [0.357]	14.18 [0.015]	11.66 [0.003]	14.72 [0.792]	16.79 [0.667]	7.87 [0.163]	10.32 [0.006]	17.04 [0.650]	21.64 [0.361]	11.03 [0.051]	

Notes: This table reports residual diagnostics for GBM, Modified GBM, GARCH, Heston, Merton, GARCH-Merton, Heston-Merton fitted to daily continuously compounded returns $R_t = \ln(S_t / S_{t-1})$ of SPY, AAPL, MSFT, AMZN. The sample is October 1, 2018 to September 30, 2019 (251 trading days). Each model is estimated from that regime and company only. Jump models use standardized residuals after the 3σ jump filter. JB is the usual Jarque-Bera normality test for the standardized residuals. Q(20) is the Ljung-Box portmanteau test for serial correlation in the standardized residuals up to 20 lags; Q²(20) is the same test on the squared standardized residuals. ARCH-LM is Engle's (1982) test for remaining ARCH effects with 5 lags. The test statistic is reported with the p-value in brackets. N/A means the diagnostic is not applicable or the residual series is not well-defined. These tests check model adequacy on returns; they are not option pricing RMSE or MAE.

Table D4 · 2019-2020 COVID · model diagnostics

2019-09-01 → 2020-08-31 · n = 252 daily observations · statistic [p-value]

Model	SPY				AAPL				MSFT					AMZN			
Model	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	JB	Q(20)	Q²(20)	ARCH-LM	
GBM	1106.07 [<0.001]	17.63 [<0.001]	70.08 [<0.001]	0.59 [<0.001]	49.69 [<0.001]	24.89 [<0.001]	33.86 [<0.001]	1.98 [<0.001]	929.74 [<0.001]	90.54 [<0.001]	262.11 [<0.001]	102.02 [<0.001]	13.31 [<0.001]	47.50 [<0.001]	14.12 [<0.001]	44.15 [<0.001]	
Modified GBM	1377.35 [<0.001]	208.08 [<0.001]	73.14 [<0.001]	8.53 [<0.001]	95.60 [<0.001]	77.72 [<0.001]	114.46 [<0.001]	7.69 [<0.001]	2099.24 [<0.001]	35.61 [<0.001]	187.91 [<0.001]	12.16 [<0.001]	243.75 [<0.001]	43.98 [0.002]	51.64 [<0.001]	30.47 [<0.001]	
GARCH	122.33 [<0.001]	9.53 [0.976]	11.89 [0.920]	3.03 [0.696]	54.88 [<0.001]	18.62 [0.547]	17.99 [0.588]	1.65 [0.895]	2.13 [0.345]	16.39 [0.692]	14.29 [0.815]	5.17 [0.395]	134.80 [<0.001]	14.14 [0.823]	11.67 [0.927]	0.40 [0.995]	
Heston	9.39 [0.009]	16.58 [0.680]	3.57 [<0.001]	5.31 [<0.001]	10.45 [0.005]	20.09 [0.452]	24.23 [0.233]	6.01 [0.305]	11.14 [0.004]	24.62 [0.216]	34.55 [0.023]	7.35 [0.196]	11.71 [0.003]	14.16 [0.822]	16.05 [0.713]	2.46 [0.783]	
Merton	1106.07 [<0.001]	17.63 [<0.001]	69.44 [<0.001]	0.54 [<0.001]	49.69 [<0.001]	24.89 [<0.001]	32.17 [<0.001]	1.60 [<0.001]	929.74 [<0.001]	90.54 [<0.001]	262.16 [<0.001]	102.14 [<0.001]	13.31 [<0.001]	47.50 [<0.001]	13.83 [<0.001]	43.96 [<0.001]	
GARCH-Me	6.48 [0.039]	18.85 [0.531]	14.68 [0.794]	3.46 [0.629]	6.54 [0.038]	19.59 [0.484]	26.18 [0.160]	8.67 [0.123]	0.49 [0.782]	17.10 [0.647]	17.27 [0.635]	11.58 [0.041]	3.15 [0.207]	19.39 [0.497]	29.61 [0.076]	12.00 [0.035]	
Heston-Me	10.09 [0.006]	12.21 [0.909]	43.10 [0.002]	7.19 [0.207]	12.19 [0.002]	18.87 [0.530]	19.70 [<0.001]	16.70 [0.244]	12.36 [0.002]	18.96 [0.524]	23.12 [0.283]	4.18 [0.524]	13.47 [0.001]	15.58 [0.742]	21.20 [0.386]	6.33 [0.276]	

Notes: This table reports residual diagnostics for GBM, Modified GBM, GARCH, Heston, Merton, GARCH-Merton, Heston-Merton fitted to daily continuously compounded returns $R_t = \ln(S_t / S_{t-1})$ of SPY, AAPL, MSFT, AMZN. The sample is September 1, 2019 to August 31, 2020 (252 trading days). Each model is estimated from that regime and company only. Jump models use standardized residuals after the 3σ jump filter. JB is the usual Jarque-Bera normality test for the standardized residuals. Q(20) is the Ljung-Box portmanteau test for serial correlation in the standardized residuals up to 20 lags; Q²(20) is the same test on the squared standardized residuals. ARCH-LM is Engle's (1982) test for remaining ARCH effects with 5 lags. The test statistic is reported with the p-value in brackets. N/A means the diagnostic is not applicable or the residual series is not well-defined. These tests check model adequacy on returns; they are not option pricing RMSE or MAE.